

DECODED EXPLAINED

26th July, 2026

Zoopharmacognosy : When Animals Medicate Themselves

Zoopharmacognosy — when animals medicate themselves



**SPEAKING OF
SCIENCE**

D. Balasubramanian

An interesting review by G. Allen et al. in *Scientific Reports* (16,18690: 2026) reports on how orangutans, which are still present in the Sebang forest in Central Kalimantan region in Indonesia, treat themselves by mixing saliva with the leaves of a plant botanists that call dracaena (a member of the asparagus family) and applying the mixture to their body in order to purify the air and relieve muscle pain and joint discomfort while carrying their babies.

Botanists from Gujarat University, Ahmedabad, had previously shown that this evergreen plant has

antioxidant, antibacterial, antifungal, and anticancer activities (Vasava et al., *Int. J. R. Tech.*, 2025; 117-120). Chimpanzees in Nigeria have also been known to swallow leaves of the plant *Desmodium gangeticum* (called 'shalaparni' in India) to get rid of parasites that bother the animal's skin. Ancient Ayurveda practitioners such as Sushruta were also known to use this plant and its parts as parts of formulations to manage asthma, haemorrhoids, and typhoid. What our ancestral primates millennia ago had practised has come down to us humans, and today we use several available plants such as snake plant, ginseng, oregano, and others for medicinal purposes.

Based on the fact that animals use plants like these to treat themselves, plant biologists have



A male Sumatran orangutan seen after self-treating a wound with a medicinal plant in the Suaq Balimbing research site in 2022. REUTERS

coined the term 'zoopharmacognosy' – 'zoo' stands for animals, 'pharmaco' for medicines, and 'gnosy' for knowledge. In the March 2008 issue of *Resonance*, the journal of the Indian Academy of Sciences, R. Raman and S. Kandula had reviewed this aspect of animals (vol. 13, pages 245-253 (2008)). Wikipedia itself lists around 70 publications in this

area: of animals self-medicating with plants, roots, and fruits.

Even common elephants have been known to chew the leaves of the oregano plant for relief from phlegm. Monkeys, sheep, dogs, and cats are also known to self-medicate by eating grass to remove parasites such as worms and amoeba that cause illness in the body.

They also consume clay or soil particles, counting on substances therein to alleviate pain. This practice is referred to as geophagy – 'geo' referring to earth and 'phagy' to consuming. Birds are also known to use herbs to protect their nests from invaders (*Phil. Trans. R. Soc. B* (2018) 373 (1751): 20170196).

What about plants? Unlike animals, plants do not have a nervous system, although they too can sense their environment, respond to insect attacks, and try to defend themselves against herbivores.

There is also evidence that monkeys share with us humans the ability to track auditory regularities. In other words they also possess a faculty to appreciate music. Y. Ayala et al., in 2017 (*Scientific Reports* 7, 16687), pointed out that the rhesus monkeys res-

pond to the acoustic syllables (musical scales), though not the rhythmic patterns as we humans do. Also in 2017, scientists at Glasgow University showed (DOI: 10.1016/j.physbeh.2017.01.024) that dogs show measurable changes in behaviour, heart rate, and stress levels when exposed to music, though the responses varied by genre, tempo, and the individual dog. On the other hand, cats have been found to respond to far different audio frequencies.

We should keep these details in mind when exposing animals to stressful environments like shelters and veterinary clinics, where the right kind of music can visibly calm anxious animals. As for friends with pet dogs: play music and let your dog enjoy it too!

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Zoopharmacognosy : When Animals Medicate Themselves

- A review by G. Allen et al. in Scientific Reports (2026) reports on how orangutans in the Sebangu forest, Central Kalimantan, Indonesia, treat themselves by mixing saliva with the leaves of a plant botanists call dracaena, and applying the mixture to their body to purify the air and relieve muscle pain and joint discomfort while carrying their babies
- Chimpanzees in Nigeria have been known to swallow leaves of *Desmodium gangeticum*, called ‘shalaparni’ in India, to get rid of parasites on their skin; ancient Ayurveda practitioners such as Sushruta also used this plant for asthma, haemorrhoids, and typhoid
- Elephants chew oregano leaves for relief from phlegm; monkeys, sheep, dogs, and cats self-medicate by eating grass to remove parasites such as worms and amoeba
- Animals also consume clay or soil particles, a practice called geophagy, to alleviate pain; birds use herbs to protect their nests from invaders

Zoopharmacognosy : When Animals Medicate Themselves

Which one of the following makes a tool with a stick to scrape insects from a hole in a tree or a log of wood?

This question was previously asked in
UPSC Civil Services Prelims
2023: General Studies (SET - A...

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1. Fishing cat
2. Orangutan
3. Otter
4. Sloth bear

UPSC Prelims 2023 Question:

Which of the following organisms perform waggle dance for others of their kin to indicate the direction and the distance to a source of their food?

1. Butterflies
2. Dragonflies
3. Honey Bees
4. Wasps

Sarnath Added to UNESCO Heritage List

Sarnath added to UNESCO Heritage List

Press Trust of India

NEW DELHI

India's ancient Buddhist site of Sarnath in Uttar Pradesh was on Saturday inscribed on the UNESCO World Heritage List. The decision was taken during the ongoing meeting of the UNESCO World Heritage Committee's 48th session in Busan, South Korea.

"New inscription on the @UNESCO #WorldHeritage List: Ancient Buddhist Site of Sarnath, #India," UNESCO posted on X. This is India's 45th site to join the list.

PM Narendra Modi hailed the inscription, stating that the site has a deep connection with Lord Buddha, whose timeless message of wisdom, compassion, and harmony inspires the world.

Sarnath Added to UNESCO Heritage List

● Sarnath : Historical Significance

- Located near Varanasi, Uttar Pradesh, Sarnath is where Gautama Buddha delivered his first sermon, the Dharmachakra Pravartana, after attaining enlightenment, making it one of Buddhism's four most sacred sites, alongside Lumbini, Bodh Gaya, and Kushinagar
- Houses the Ashokan Pillar with the Lion Capital, India's national emblem, and the Dhamek Stupa, among other ancient Buddhist monuments

● UNESCO World Heritage List

- A list of sites of outstanding universal cultural or natural significance, maintained under the 1972 World Heritage Convention, to which India is a signatory
- Inscription requires meeting at least one of ten selection criteria related to cultural or natural heritage value, along with authenticity, integrity, and an adequate protection and management system

● India's World Heritage Sites

- With Sarnath's inclusion, India now has 45 UNESCO World Heritage Sites, spanning cultural sites, natural sites, and mixed sites
- Other Buddhist heritage inscriptions include the Mahabodhi Temple Complex at Bodh Gaya and the Buddhist Monuments at Sanchi

● Ashoka and the Spread of Buddhism

- Emperor Ashoka, in the 3rd century BCE, played a central role in patronising and spreading Buddhism, erecting pillars and stupas at key sites including Sarnath after his conversion following the Kalinga war

Sarnath Added to UNESCO Heritage List

UNESCO World Heritage Sites are landmarks of outstanding cultural, historical, or natural significance protected under an international treaty administered by the United Nations Educational, Scientific and Cultural Organisation. India, with its civilisational depth and ecological diversity, is home to 45 such sites

Properties on World Heritage List (44)

Cultural World Heritage Properties in India	:	37
Natural World Heritage Properties in India	:	07
Mixed World Heritage Properties in India	:	01

India's Rice Fields Rank Among World's Top Methane Sources

India's rice fields rank among world's top methane sources

India is the world's top rice producer, with its area under paddy cultivation covering 55% of its rainfed harvested area; continuous irrigation, incorporating residues and increasing use of nitrogen fertilizers has increased emissions

Sharmila Vaidyanathan

Greenhouse gas (GHG) emissions from paddy fields have doubled since the 1960s, a new study in *Nature Food* has found. In 2011-2020, paddy fields worldwide produced around 1.1 billion tonnes of carbon dioxide-equivalent (CO₂-e) emissions every year, up 90.4% from annual emissions in 1961-1980.

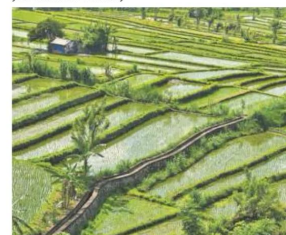
Per the study, around half the increase was driven by the loss of soil organic carbon. In 1961-1980, paddy fields removed 289 million tonnes of CO₂-e emissions annually. But by the 2010s, over a third of paddy fields worldwide had begun losing carbon.

Methane from paddy fields was also a significant driver. Flooded fields develop low-oxygen conditions where methane-producing microbes thrive. And paddy fields in eastern India were among the global methane hotspots.

"Most previous studies have focused on a single greenhouse gas, a specific management practice or a particular region," Hanqin Tian, co-author of the study and director and institute professor, Department of Earth and Environmental Sciences, Boston College, the U.S., said. "Our study provides a much more comprehensive assessment by integrating a global meta-analysis of more than 1,200 field experiments, process-based ecosystem modelling, and artificial intelligence approaches to quantify GHG emissions from rice cultivation worldwide."

Dirty rice

India's irrigated paddy fields release more methane in a year than all of Germany



Rice is crucial for India's food security. STEVE DOUGLAS/UNSPLASH

India is the world's top rice producer. Its area under paddy cultivation has increased by around 25% since the 1970s, currently covering 55% of India's rainfed harvested area. This together with continuous irrigation, incorporating residues such as leftover stalks, leaves, and stubble, and increasing use of nitrogen fertilizers, has increased emissions.

"Our analysis indicates that a combination of improved irrigation, optimised fertilizer use, residue management, and reduced tillage could reduce global rice-related greenhouse gas emissions by about 10% by mid-century, even under a warming climate," Dr. Tian said.

The study also projected that between 2031 and 2050, annual emissions will increase by a quarter, fuelled by global warming, erratic rainfall, and the intense use of agricultural inputs. Irrigated paddy fields

in India already release 3.9 million tonnes of methane a year, a little more than the annual methane emissions of Germany.

While India has not set up goals to reduce emissions from agriculture, some techniques and incentives are available to farmers. The National Innovations in Climate Resilient Agriculture (NICRA) project suggests – among others – the 'System for Rice Intensification' technique: young seedlings are transplanted precisely and fields are not continuously flooded during irrigation, reducing emissions.

However, the methods don't address the emission reduction potentials varying widely between States. Apart from demonstrations of alternatives across districts under NICRA, several national and State-level policies and incentives are designed to encourage farmers to adopt the alternative methods. However,

- Global greenhouse gas emissions from paddy fields have nearly doubled since the 1960s, a new study has found

- Half of the emissions increase stems from the loss of soil organic carbon in paddy fields worldwide

- High methane levels result from flooded fields, continuous irrigation, and heavy use of nitrogen fertilizers

- Better irrigation and residue management could reduce global rice emissions by 10% by the year 2050

- However, adopting new techniques is often constrained by labour shortages and a lack of technical knowledge among farmers

- Crop diversification and limiting paddy expansion are essential to meeting regional environmental and climate goals

identifying low-emission packages, where a combination of these factors can help reduce emissions for various paddy varieties," Dr. Veertil said.

He shared the example of lime that farmers use to neutralise the State's acidic soil. "We have observed that replacing a portion of the lime with phosphogypsum has a positive impact on emissions in the subsoil surface. We are also experimenting with different formulations and application methods for nitrogen fertilizers."

Ashwini Chhatre, associate professor of public policy and executive director at the Bharti Institute of Public Policy, Hyderabad, said that depending on a region's suitability for paddy, farmers should have support to opt for low-emission techniques or grow alternative local crop varieties. In his research, Dr. Chhatre found that farmers still dedicate large areas of their farmlands to paddy. "As a general strategy, there is no way forward without crop diversification," he said. "Despite the agro-climatic diversity in the country, paddy has the largest area under cultivation at the cost of regional indigenous crops."

"An integrated intervention for alternative grains is crucial," Dr. Chhatre continued, adding that it "must include" extension and training services, involve self-help groups, and provide procurement support. "Policies must also prevent further expansion of paddy fields to curb emissions."

(Sharmila Vaidyanathan is an independent journalist from Bengaluru)

India's Rice Fields Rank Among World's Top Methane Sources

Rice

Rice is mainly a Kharif crop (sown in June–July and harvested in Nov–Dec), but it is also grown as Rabi and Zaid crops in Southern and Eastern India.

Climate: Rice requires temperatures of 25–35°C, high humidity, and annual rainfall above 100 cm.

Soil: It grows best in deep, clayey, and loamy soils with high water-holding capacity.

Rice Production in India

Global Share: India produced a record 150.18 MT in 2024-25, accounting for over 28% of global output.

Area Coverage: Rice occupies largest cropped area in India, covering about 51–52 million hectares.

Key States: West Bengal is the leading producer (about 15%), followed by Uttar Pradesh (~12%) and Punjab (~10%).

Punjab has the highest yield due to full irrigation and mechanisation, followed by Tamil Nadu and Telangana.

Export: India is the world's largest rice exporter, accounting for nearly 40% of global trade.

Basmati rice is exported to West Asia; non-Basmati rice is exported to Africa and Southeast Asia.

India's Rice Fields Rank Among World's Top Methane Sources

● The Study Findings

- Greenhouse gas emissions from paddy fields have doubled since the 1960s, according to a new study in Nature Food; in 2011-2020, paddy fields worldwide produced around 1.1 billion tonnes of CO2-equivalent emissions annually, up 90.4% from 1961-1980 levels

- High methane levels result from flooded fields, continuous irrigation, and heavy use of nitrogen fertilizers; paddy fields in eastern India were among the global methane hotspots

74 percent of the nitrous oxide emissions over the last decade originated from agricultural practices, specifically the use of nitrogen fertilizers and animal manure..

● India's Specific Position

- India is the world's top rice producer; its area under paddy cultivation has increased by around 25% since the 1970s, currently covering 55% of India's rainfed harvested area

● Solutions and Constraints

- The study projects that better irrigation and residue management could reduce global rice emissions by 10% by 2050, even under a warming climate

- The National Innovations in Climate Resilient Agriculture (NICRA) project suggests, among others, the 'System for Rice Intensification' technique, where young seedlings are transplanted precisely and fields are not continuously flooded, reducing emissions

- However, adopting new techniques is often constrained by labour shortages and a lack of technical knowledge among farmers; the methods also don't address how emission reduction potential varies widely between states

India's Rice Fields Rank Among World's Top Methane Sources

- **Methane as a Greenhouse Gas**

- Methane is a potent greenhouse gas with a global warming potential many times that of CO₂ over a 20-year period, though it has a shorter atmospheric lifetime; agriculture, especially rice cultivation and livestock, is a major anthropogenic source

- **Why Flooded Rice Fields Produce Methane**

- Continuous flooding creates low-oxygen, anaerobic conditions in paddy soil, ideal for methane-producing microorganisms that decompose organic matter

- **Alternate Wetting and Drying / System of Rice Intensification**

- Water-saving irrigation techniques that periodically dry out paddy fields instead of continuous flooding, reducing methane emissions while maintaining or improving yields

India's Rice Fields Rank Among World's Top Methane Sources

- **System of Rice Intensification**

- Developed first in Madagascar by Henri de Laulanie
- System of Rice Intensification (SRI) as one of the best practices for sustainable land and ecosystem management (SLEM) under the World Bank funded SLEM Project.
- Rice is one of the high-water demanding crops in India. One kilogram of rice production needs about 3000-5000 litres of water
- SRI involves cultivating rice with as much organic manure as possible, starting with young seedlings planted singly at wider spacing in a square pattern; and with intermittent irrigation that keeps the soil moist but not inundated, and frequent inter-cultivation with a weeder that actively aerates the soil.

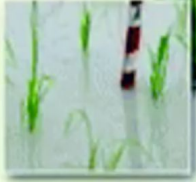


India's Rice Fields Rank Among World's Top Methane Sources

- Decrease water consumption by almost 40%
- Increase yields by over 30%
- Potential to double income level of farmers by enhancing input use efficiency and higher productivity
- Can easily be adopted by Small and Marginal farmers who account for 83% of farmers
- Greater tolerance to abiotic (drought, heat waves) and biotic (pest and diseases) stresses
- Promote environment friendly practices such as Organic farming, reduce emission of methane, improvement in soil fertility etc.
- Potential to apply this method for other crops such as Sugarcane, wheat, millet etc.

The infographic illustrates six sustainable rice farming practices, each with a numbered title and a corresponding photograph:

- 1** Seedlings get transplanted at a much younger age. 
- 2** Only single seedlings, instead of a handful of seedlings get planted in each hill. 

- 3** Plants are spaced wider apart, and in a square pattern. 
- 4** Intermittent water application to create wet and dry soil conditions, instead of continuous flood irrigation. 

- 5** Rotary weeding to control weeds and promote soil aeration. 
- 6** Increased use of organic fertilizer to enhance soil fertility. 

What Caused the Tunnel Disaster in Sikkim's Namchi?

■ Cultivation Techniques:

- **Traditional Transplantation:** Seeds sown in a nursery; seedlings transplanted to the main field after 25–35 days. Labour and water-intensive, requiring around 25–27 irrigations, but maximizes yield. This is widely practiced across India.
- **Direct Seeding of Rice (DSR):** Pre-germinated seeds are directly drilled into the field using a machine that saves water and labour. Best suited for heavy/medium-textured soils (high clay content) with good water retention and rich in plant-available iron.

What Caused the Tunnel Disaster in Sikkim's Namchi?

UPSC Prelims 2022 Question:

"System of Rice intensification" of cultivation, in which alternate wetting and drying of rice field is practised, results in

1. Reduced seed requirement
2. Reduced methane production
3. Reduced electricity consumption

Select the correct answer using the code given below:

1. 1 and 2 only

2. 2 and 3 only

3. 1 and 3 only

4. 1, 2 and 3

What Caused the Tunnel Disaster in Sikkim's Namchi?

Among the following, which one is the largest exporter of rice in the world in the last five years?

This question was previously asked in
Official UPSC Civil Services Exam
2019 Prelims Part A

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1. China
2. India
3. Myanmar
4. Vietnam

What Caused the Tunnel Disaster in Sikkim's Namchi?

What caused the tunnel disaster in Sikkim's Namchi?

How did 25 people die? Why is methane trapped in mountain rock? Why are the Himalayas fragile?

Rahul Karmakar

The story so far:

Two gas leaks, generated by a sudden explosive release of suspected methane trapped or embedded in rocks on July 20, claimed the lives of 23 workers (18 men and 5 women) and injured 2 others while constructing the Tista Valley Tunnel (TVT) of the Sikkim, 300 MW Teesta Stage V Hydropower Project at Samundring in Sikkim's Namchi district.

How did the tragedy happen?

The NDC has said a thorough investigation will be conducted to ascertain the cause of the incident. The Sikkim government has also announced the constitution of a high-level committee for the same purpose. Preliminary reports indicate the formation of toxic gases in the THT owing to the sudden release of suspected methane trapped in the rocks, resulting in an explosion and partial damage to the tunnel. This indicates that the workers hit with rock while drilling or constructing the THT, a critical component of a hydropower plant for transporting massive volumes of water from the primary intake at a reservoir to the power generation turbines.

NDC officials said that surveys for tunnel projects in India, especially in the fragile Himalayan belt, are conducted across planning, construction, and maintenance phases to ensure structural stability, worker safety, and environmental compliance.

Methane deposits are found in the Himalayan belt or other mountainous regions with active or historical coal-bearing strata, where ancient organic matter has decomposed over millions of years. These typically involve comprehensive topographical, geological, geophysical, hydrological, and geotechnical surveys, and regular monitoring and safety audits.

However, no test can detect the presence of the colorless and odorless methane, an extremely flammable gas, which can explode after coming in contact with air in a specific ratio and encountering an ignition source, possibly sparks during drilling in the case of the Teesta-VI tunnel.

How is methane trapped in hilly terrain?

Methane deposits usually occur either as natural gas trapped in porous rock or so-called gas pockets in young, fragile geological formations. These are frequently encountered in the Himalayan belt or other mountainous regions with active or historical coal-bearing strata, where ancient organic matter has decomposed over millions of years. Hilly depressions or valleys frequently trap moisture, creating oxygen-depleted (anaerobic) waterlogged soils, where microscopic organisms digest organic matter and emit methane. Improperly managed or open waste dumps in hilly terrain also decompose in anaerobic layers, producing large amounts of landfill gas.

Moreover, hilly and mountainous regions are often located near fault lines or areas with underlying coal, gas, or geothermal reserves, with natural fissures allowing the migration of underground methane deposits. There are no coalfields in Sikkim, but the State is in the Rangit Tectonic Window zone, and geological surveys of India studies have identified coal reserves of approximately 1.4 billion tonnes in southern Sikkim, primarily the Namchi area where the NDC tunnel incident occurred.

According to the State Mines and Geology Department, a regional-level coal exploration project commenced in the district in December 2020 under a Government of India programme. The coal in Namchi was sub-bituminous, containing 82.40% fixed carbon, 20.5% ash, 3.1% volatile matter, and 3.7% moisture.

Have other prospects in Sikkim been similarly affected?

A young geological age, active tectonic instability, and steep, erosion-prone terrain make the Himalayan fragile. The mountains are constantly rising and shifting due to the collision of the Indian and Eurasian plates, making the region highly prone to earthquakes. Heavy Monsoon and landslides on highways, dams, and railways add to the geological instability, leading to frequent landslides and mudslides. Heavy tourism, unregulated urban expansion, poor waste management, and deforestation have compounded the problems.

Sikkim has also suffered from extreme weather events such as cloudbursts, rapidly melting glaciers, and expanding glacial lakes, attributed to climate change. Hydropower projects in the State have suffered owing to several of these factors.

A Glacial Lake Outburst Flood in October 2018 killed more than 90 people and destroyed Sikkim's 100 km-long 1,400 MW dam at Chungthang in Margat district. The disaster also damaged NDC's 300 MW Teesta-V Hydropower Project, which was resumed three years later. A landslide damaged some houses and a building of this project in August 2024.

The Teesta-VI tunnel disaster is the first recorded incident in the State involving an explosion due to suspected methane gas release.

What Caused the Tunnel Disaster in Sikkim's Namchi?

The Geological Survey of India (GSI), one of the oldest scientific organizations in the country, is set to celebrate its 175th year of geoscientific legacy

It was set up in 1851 by Sir Thomas Oldham, primarily to find coal deposits for the Railways.

Over the years, it has grown into a repository of geoscience information required in various fields in the country.

Headquarter: It is headquartered in Kolkata and has six regional offices located at Lucknow, Jaipur, Nagpur, Hyderabad, Shillong and Kolkata. Every state has a state unit.

Nodal Ministry: Presently, GSI is an attached office to the Ministry of Mines.

Why Are Mega Data Centre Projects Facing Stiff Resistance?

Why are mega data centre projects facing stiff resistance?

EXPLAINER

Paul Nicodemus
Snehal Mutha

This week, people in Andhra Pradesh and Maharashtra took to the streets to oppose the proposed mega data centre projects in the States.

In A.P., after the government allotted about 601 acres of land across three locations in Visakhapatnam and Anakapalli district, Raiden Infotech India Pvt. Ltd. – a Google company – began developing a 1GW AI data centre hub, with Adani Infra/Adani-Connex as the notified development partner.

In Maharashtra, residents of Thane have been opposing Amazon's proposed hyperscale data centre project. The ₹7,203.29-crore project spread over 53.73 acres is under construction across

the villages of Balkum Pada and Majiwada in Thane.

Reasons for opposition

Citizens' groups in Visakhapatnam have opposed the Andhra project, highlighting concerns over the strain on resources such as water and power, as well as its environmental impact. The project is coming up at three locations – Adavivaram-Mudasarlova, Tarluwada, and Rambilli.

Campaigners also cited that the Super El Niño-like weather pattern has put strain on water resources across the State.

In Thane, people have cited sketchy details, discrepancies in water usage data, lack of consultation with locals as reasons to rally against the project.

Residents represented by the citizen group 'Wakeup Thanekar' has demanded clarity on the water usage and a public hearing. It has sought the



Citizen groups hold a protest against proposed data centre projects in and around Visakhapatnam, along Beach Road on July 19. V. RAJU

establishment of baseline noise and air quality measurements, and a clarity on the actual benefit to locals.

The opposition from citizens was not to technology or investment but to a project that offered little in return for its costs, said Sohan Hatangadi, a campaigner in Visakhapatnam. He cited limited permanent employment and limited ancillary industry, even as the city provides

land and incentives.

President of Green Visakha, Raja Rama Mohan Roy, said the project risked the "reckless exploitation of nature". He said the Kambalakonda Eco-Forest, and the surroundings of the Rambilli Wildlife Sanctuary faced a threat, along with the Mudasarlova reservoir, a source of drinking water for the city.

In Thane, activists are against "unplanned deve-

lopment". "We cannot have the data centre in residential area, which are already suffering water shortage, load shedding, and other issues," Sadhana Pradhan, convenor of Wake-up Thanekar said.

The group plans to file a petition with the National Green Tribunal after the government response. It had held a meeting with the Thane municipal commissioner and representatives of Amazon. "We are disappointed as they had no answers to basic question of what will be the source of the water required for cooling," Ms. Pradhan said.

Concerns over water

A more pressing constraint for Visakhapatnam was inadequate storage capacity, a concern that engineers had flagged internally, D. Murali Krishna, Executive Engineer, Water Supply, of the Greater Visakhapat-

nam Municipal Corporation said. He said roughly 1,000 to 2,000 acres are needed to be set aside for reservoirs, and that around 5,000 million cubic feet of storage could sustain the city for nearly six months.

Meanwhile, the Thane project will require 422 MW power and backup of 193 generators. Wakeup Thanekar group said that Amazon and the Thane corporation have given two different figures on water usage. While Amazon claimed 125 kilolitres a day per day usage, the corporation announced that it will provide 12 million litres a day of recycled sewage water.

Jatin Kothare, a resident in Balkum, said that the city is already "short of 40 MLD water".

"So how you going to fulfil the water requirements of the data centre?" he asked.

Why Are Mega Data Centre Projects Facing Stiff Resistance?

● Data Centres and the AI Infrastructure Boom

○ Data centres house the servers and computing infrastructure that power cloud computing, AI model training, and digital services; the global AI boom has driven a surge in hyperscale data centre construction, requiring massive amounts of electricity and water for cooling

● Water Usage in Data Centres

○ Large data centres use significant water for cooling systems; water-stressed regions face particular strain, prompting demands for use of recycled and treated water and transparent usage disclosures

● India's Push to Become a Global Data Centre Hub

○ States have been competing to attract data centre and AI infrastructure investment through land allotments and incentives, seeing it as a driver of digital economy growth, even as local resource and environmental concerns mount

Why Are There Protests Over the Ken-Betwa Project?

Why are there protests over the Ken-Betwa project?

Why does the project aim to do? How many villages and families are likely to be affected by it? What are the concerns of the tribal people regarding rehabilitation and compensation? Why has the project raised concerns among conservationists?

Mehul Malpani

The story so far:

Madhya Pradesh police personnel went to the banks of the Barana river stream in Kupi village of Chhatarpur district on July 19 and cleared a protest site where hundreds of villagers, mostly tribal people, had been demonstrating for two weeks against the Union government's ₹44,000 crore Ken-Betwa Link Project, India's first river-linking project, and two irrigation projects of the State government.

The police took local activist Amit Bhatnagar, founder of the Jai Kisan Sangathan, who had been on an indefinite hunger strike for 14 days, and admitted him to a government hospital. They also picked up nearly 150 protestors, most of them from neighbouring Panna district, in buses and dropped them off at their villages.

On July 23, he suspended his 18-day fast after the government's assurance that people left out of the rehabilitation process will be included.

What is the Ken-Betwa river linking project? The Ken-Betwa River Linking Project is the first of 30 proposed river-linking projects under the National Perspective Plan (NPP). Formulated in

Protesters have alleged irregularities in the gram sabha proceedings in tribal villages, claiming that the process of securing consent for the Ken-Betwa project was flawed

1980 by the then Union Ministry of Irrigation and the Central Water Commission, the NPP aims to transfer water from rivers with a perceived surplus to those facing water deficits. The NPP is divided into two components – Himalayan Rivers Development that proposes 14 links and Peninsular Rivers Development that plans 16 links, as identified by the National Water Development Agency.

The foundation stone for the project was laid by Prime Minister Narendra Modi in December 2024, in Khajuraho, where he also laid the foundation stone for the construction of the Daudhan Dam during a public event.

The project aims to transfer the declared surplus water of the Ken River to the Betwa River by constructing the Daudhan Dam on the Ken and a 221-kilometre canal, including a 2-km tunnel. In addition, the project is expected to generate 103 MW of hydropower and 27 MW of solar power. It also aims to address water scarcity in the drought-prone Bundelkhand region by providing drinking water to about 4.4 million people across 12 districts of Madhya Pradesh and more than 2 million people across 10 districts of Uttar Pradesh.

Why are tribal people protesting and what are their key demands?

According to the Union government, 7,193 families of 14 villages in Chhatarpur and eight villages in Panna district are facing displacement under the project. According to Divya Ahirwar of Jai Kisan Sangathan, the key demands regarding the Ken-Betwa project are that each adult be treated as an individual beneficiary – rather than as part of a family – for the ₹12.5 lakh rehabilitation package, and that the eligibility cut-off date be revised from February 16, 2024, to the present date.

The protests also oppose the State government's Majhgaon and Runjh irrigation projects. For these projects, the protesters are demanding rehabilitation and compensation on par with the package offered under the Ken-Betwa project. In both cases, they have also alleged serious irregularities in the survey and

selection process and claim that several people who have no connection with the villages or have moved years ago have been included in the rehabilitation plan, while many genuine residents have been left out.

Speaking from the hospital, Mr. Bhatnagar claimed that the police action came on the day he was planning to "reveal the details of an alleged ₹400-crore corruption scheme" involving the inclusion of fake beneficiaries in the compensation process.

Protesters have also alleged irregularities in the gram sabha proceedings in tribal villages, claiming that the process of securing consent for the Ken-Betwa project was flawed.

How did the government respond?

The State government has approved an additional ₹300 crore for the rehabilitation packages of the Majhgaon and Runjh irrigation projects, increasing compensation from ₹5 lakh to ₹12.5 lakh per eligible beneficiary.

Chhatarpur Collector Parth Jaiswal said that a new survey had identified around 750 additional families, who have now been included in the rehabilitation plan for the Majhgaon and Runjh projects. He described the protesters' remaining demands as "invalid" and alleged that some people were "misleading" the villagers.

What are the environmental concerns surrounding the project?

Several environmental concerns have also been flagged, especially as a large part of the project falls inside the Panna Tiger Reserve. Apart from displacement, the primary concern revolves around the proposed felling of trees. As per the Jal Shakti Ministry, approximately 17,101 trees had been identified for felling to facilitate the execution of the Daudhan Dam and allied infrastructure until July, 2025. Of these, 12,404 trees have been felled within the Reserve Forest.

There is worry that the project may hurt the tiger reintroduction programme in the reserve which revived the feline population after it had locally gone extinct in 2009. Apart from tigers, species such as endangered vultures, mahseer fish, and gharials in the Ken Gharial Sanctuary are also feared to be impacted.

The Central government is yet to release hydrological data of the two basins claiming that they are sensitive by virtue of being subsets of the international Ganga basin.



The protest against the Ken-Betwa River Linking Project and other development projects, in Chhatarpur district, Madhya Pradesh. PTI

Why Are There Protests Over the Ken-Betwa Project?

- **National Perspective Plan and River Linking**

- Formulated in 1980 by the Union Ministry of Irrigation and the Central Water Commission, the National Perspective Plan aims to transfer water from rivers with a perceived surplus to water-deficit regions, divided into Himalayan Rivers Development, with 14 links, and Peninsular Rivers Development, with 16 links, identified by the National Water Development Agency

- **Panna Tiger Reserve**

- A tiger reserve in Madhya Pradesh where tigers had gone locally extinct by 2009 due to poaching, later revived through a reintroduction programme, now a conservation success story that campaigners fear could be jeopardised by the Ken-Betwa project

Thank You